

Project Proposal: A Website for Personalized Dietary Assessment and AI-Assisted Calorie Tracking

Workshop L3 - Computer Science (SCI)

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1. Introduction

Obesity is a major health problem that is not only related to food intake but also influenced by hormonal imbalances, metabolic issues, lifestyle factors, psychological stress, and chronic conditions. People who are overweight often struggle to lose weight because they cannot accurately estimate calorie intake, track their diet, or understand how factors such as gut health, insulin resistance, or thyroid disorders affect their progress. Additionally, managing changing habits, adopting a healthier lifestyle, and losing weight can be challenging without proper guidance and monitoring.

2. Objective

With the rapid advancement of web technologies and artificial intelligence, intelligent digital platforms offer new opportunities to support dietary monitoring and personalized nutrition. This project aims **to design and implement a web-based platform for online personalized dietary consultation and AI-assisted calorie tracking**. The platform enables users to access worldwide online nutrition consultations, follow personalized dietary plans based on their health history, and track their progress under professional supervision. In addition, an AI-based food recognition system is integrated to estimate meal calories, supporting users in increasing food awareness and maintaining healthy habits, particularly outside structured diet plans.

While personalized consultations remain the core service, the AI calorie estimation feature enhances autonomy, awareness, and long-term habit building.

In addition to personalized consultation and AI-assisted calorie tracking, the platform provides structured payment and subscription systems. Each nutrition plan has a defined duration, associated price, and specific follow-up conditions, and users must complete payment to access personalized consultations. Seasonal or limited-period plans are also offered, such as summer programs or special plans for Ramadan, providing flexible options to suit users' needs throughout the year.

The platform also integrates content and community features to enrich user engagement. Blog and news sections provide educational resources on nutrition, healthy recipes, and general lifestyle

and wellness insights. Users can share personal experiences, provide testimonials, and subscribe to newsletters for updates, fostering a supportive and interactive community alongside the core nutritional services.

3. Actors and Roles

The platform involves several key actors, each with specific roles and responsibilities:

3.1 User (Patient / Client)

The user is the primary beneficiary of the platform. It is designed to support a wide range of individuals, including:

- People struggling with weight loss resistance
- Individuals managing insulin resistance, polycystic ovary syndrome (PCOS), thyroid disorders, and diabetes
- Mothers navigating postpartum recovery and body confidence
- Young professionals experiencing bloating, chronic fatigue, and burnout
- Women over 40 seeking to balance hormones, age healthily, and maintain vitality
- Individuals seeking to break free from repeated dieting cycles and adopt a healthier, more sustainable lifestyle

Users can:

- Register and log in to the website
- Manage personal profile information and health history
- Book and pay online nutrition consultations worldwide through zoom calls.
- Follow personalized dietary plans based on their medical history and goals
- Track food intake using AI-assisted calorie estimation
- Monitor progress and receive guidance and follow-up
- Access educational content such as blogs and nutrition news
- Share personal experiences and feedback
- Contact the administrator when necessary

3.2 Nutritionist

The nutritionist acts as both a clinical nutrition expert and a health transformation coach. Their role is to assess patients' medical history, lifestyle, and dietary habits in order to design personalized nutrition plans adapted to individual needs and health conditions. These plans take into account factors such as weight loss resistance, hormonal imbalances, metabolic disorders, and chronic diseases. Their role includes to:

- Conduct online consultations
- Design personalized nutrition plans
- Monitor user progress and adjusting plans when necessary
- Communicate with users for follow-up and support

3.3 Administrator

The administrator ensures the proper functioning and management of the platform. Responsibilities include:

- Managing users, nutritionists, and subscriptions
- Overseeing content such as blogs and announcements
- Handling user inquiries and platform maintenance
- Ensuring data security and system reliability

3.4 AI-Based Calorie Estimation

The AI-based calorie estimation feature is a key functional component of the platform, designed to support users in monitoring their daily food intake. This feature complements professional nutrition consultations and personalized dietary plans, rather than replacing expert guidance.

The platform uses computer vision and deep learning to recognize food and estimate calories. The system analyzes images of meals, identifies the food items, and estimates portion sizes. Calorie values are then calculated using standard nutritional databases.

Functionality

- The user uploads an image of a prepared meal
- A deep learning model performs food recognition using object detection and segmentation
- Identified food items are matched with nutritional references
- The system estimates the calorie content of the meal
- Results are presented as approximate values for self-monitoring and awareness

Use Cases The AI-based calorie estimation feature is primarily intended for users who are not currently following a personalized nutrition plan, those enrolled in seasonal or flexible programs, and individuals seeking to maintain their weight after completing a structured nutrition plan.

This AI-assisted approach increases food awareness, supports habit building, and encourages long-term lifestyle management while acknowledging the inherent uncertainty of visual calorie estimation.

4. Conclusion

This project proposes an online, worldwide nutrition management platform focused on personalized consultations and AI-assisted calorie tracking. By combining professional supervision, intelligent food analysis, and accessible digital services, the platform aims to support individuals facing complex weight-related challenges and help them adopt sustainable, healthy lifestyles. As part of the academic component, the student will conduct research on vision-based nutritional assessment systems, compare the results with existing literature, and document the methodology, experiments, and findings in the final dissertation. This approach not only contributes to the scientific understanding of AI-assisted dietary monitoring but also delivers a practical, user-centered solution for modern healthcare challenges.

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